

11(b). Weighting Estimators

ISS5096 || ECI

Jaewon (“Jay-one”) Yoo

National Tsing Hua University

Roadmap

1. Weighting Estimators

2. Looking Ahead: From AIPW to DML

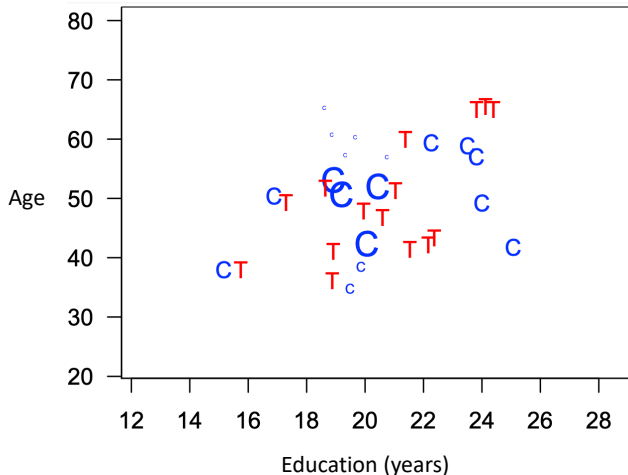
Where are we? Where are we going?

- The recurring question of this course: **where do we find good controls?**
 - Units *randomized* to receive control (RCT)
 - Units with *similar* values of covariates (SOO / reg. adj.)
 - Units with *opposite* value of some instrument (IV)
 - Units *near a discontinuity* in treatment assignment (RDD, W13)
 - Exploit two possible sources of variation for identification:
 - Exploit **cross-sectional** variation in treatment.
 - Exploit variation in treatment **within a unit over time** (before/after)
- Can we make our identification strategies work *better*?
 - $\rightsquigarrow$ **matching** or **weighting**
 - Today: refine the SOO branch by pruning or re-weighting observational controls.

1/ Weighting Estimators

Weighting vs. Pruning, Visually

- Same propensity score, two complementary uses: **prune** (matching) or **re-weight** (IPW).



Credit: Figure from Gary King's talk on "The Balance-Sample Size Frontier in Matching Methods for Causal Inference," at University of Michigan, January 24, 2014.

Why Weighting?

- Matching has a couple of downsides:
 - Inefficient: it may throw away data.
 - Ineffective: crude tool so it may not be able to balance covariates.
- Matching is actually a *special case* of a weighting estimator:

$$\begin{aligned}\hat{\tau}_m &= \frac{1}{n_1} \sum_{i=1}^n D_i \left(Y_i - \frac{1}{M} \sum_{j \in \mathcal{J}_M(i)} Y_j \right) \\ &= \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} \underbrace{\left(\frac{n_0}{n_1} \frac{K_M(i)}{M} \right)}_{\text{weight}} Y_i\end{aligned}$$

- $K_M(i)$ is the number of times i is used as a match.
- Weighting estimators choose the weights directly to reduce imbalance.

Survey Sampling: Where IPW Comes From

- Imagine we want to estimate the population mean $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$
 - $R_i = 1$ if sampled, $R_i = 0$ if not.
 - We only observe Y_i for those with $R_i = 1$.
 - Inclusion probability varies by person: $\mathbb{P}(R_i = 1) = \pi_i$
- **Horvitz-Thompson estimator** is unbiased (treating Y_i as fixed):

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \frac{R_i Y_i}{\pi_i} \right] = \frac{1}{n} \sum_{i=1}^n \frac{\mathbb{E}[R_i] Y_i}{\pi_i} = \frac{1}{n} \sum_{i=1}^n \frac{\pi_i Y_i}{\pi_i} = \bar{Y}$$

- Key idea: **reweight** the observed sample to be representative of the population we cannot fully observe.

Horvitz-Thompson for Treatment Effects

- Applying HT to potential outcomes: weight by inverse propensity score.

$$\widehat{ATE} = \widehat{\tau}_{ipw} = \frac{1}{n} \sum_{i=1}^n \left(\frac{D_i Y_i}{\widehat{\pi}(\mathbf{X}_i)} - \frac{(1 - D_i) Y_i}{1 - \widehat{\pi}(\mathbf{X}_i)} \right)$$

- Under no unmeasured confounders, $\mathbb{E}[\widehat{\tau}_{ipw}] \xrightarrow{P} \tau$ (consistent)
 - Would be unbiased if we knew the true propensity scores, $\pi(\mathbf{X}_i)$
- Similar expression for ATT:

$$\widehat{ATT} = \widehat{\tau}_{ipw,t} = \frac{1}{n_1} \sum_{i=1}^n \left(D_i Y_i - \frac{\widehat{\pi}(\mathbf{X}_i)(1 - D_i) Y_i}{1 - \widehat{\pi}(\mathbf{X}_i)} \right)$$

- Logic: up-weight units with “rare” treatment values for their values of $\mathbf{X}_i$
 - A kind of “continuous” version of matching with replacement.

Normalized Weights (Hajek)

- HT estimators with known weights are unbiased but can be *inefficient*.
 - Large weights can lead to highly variable estimates when (not) included.
- Hajek estimator** normalizes the denominator so the weights sum to 1:

$$\hat{\mu}_h = \hat{\mathbb{E}}[Y_i] = \frac{\sum_{i=1}^n (R_i/\pi_i) Y_i}{\sum_{i=1}^n R_i/\pi_i}$$

- Hajek estimator for the ATE:

$$\hat{\tau}_h = \frac{\sum_{i=1}^n D_i Y_i / \hat{\pi}(\mathbf{X}_i)}{\sum_{i=1}^n D_i / \hat{\pi}(\mathbf{X}_i)} - \frac{\sum_{i=1}^n (1 - D_i) Y_i / (1 - \hat{\pi}(\mathbf{X}_i))}{\sum_{i=1}^n (1 - D_i) / (1 - \hat{\pi}(\mathbf{X}_i))}$$

- Practically, **weighted least squares** gives automatic normalization:

$$(\hat{\alpha}_{\text{wls}}, \hat{\tau}_{\text{wls}}) = \arg \min_{\alpha, \tau} \sum_{i=1}^n \left(\frac{D_i}{\hat{\pi}(\mathbf{X}_i)} + \frac{1 - D_i}{1 - \hat{\pi}(\mathbf{X}_i)} \right) (Y_i - \alpha - \tau D_i)^2$$

Variance of IPW Estimators

- If $\widehat{\pi}(\mathbf{X}_i)$ is estimated, how to estimate $\mathbb{V}[\widehat{\tau}_{ipw}]$ or $\mathbb{V}[\widehat{\tau}_h]$?
- First option: **bootstrap** (possibly with clustering if needed)
- Second option: **method of moments** (Newey and McFadden, 1994)
 - Treat this as a joint estimation problem and use the delta method.
 - Moment conditions for the propensity score model with parameters θ :

$$\mathbb{E} \left[\underbrace{\left(\frac{D_i}{\pi_{\theta}(\mathbf{X}_i)} - \frac{1 - D_i}{1 - \pi_{\theta}(\mathbf{X}_i)} \right) \frac{\partial \pi_{\theta}(\mathbf{X}_i)}{\partial \theta}}_{\text{score for treatment model}} \right] = 0$$

- Moment conditions for weighting estimators:

$$\text{HT: } \mathbb{E} \left[\frac{D_i Y_i}{\pi_{\theta}(\mathbf{X}_i)} - \mathbb{E}[Y_i(1)] \right] = \mathbb{E} \left[\frac{(1 - D_i) Y_i}{1 - \pi_{\theta}(\mathbf{X}_i)} - \mathbb{E}[Y_i(0)] \right] = 0$$

$$\text{Hajek: } \mathbb{E} \left[\frac{D_i (Y_i - \mathbb{E}[Y_i(1)])}{\pi_{\theta}(\mathbf{X}_i)} \right] = \mathbb{E} \left[\frac{(1 - D_i) (Y_i - \mathbb{E}[Y_i(0)])}{1 - \pi_{\theta}(\mathbf{X}_i)} \right] = 0$$

- Replace with sample versions and use delta method to get asymptotic variance.

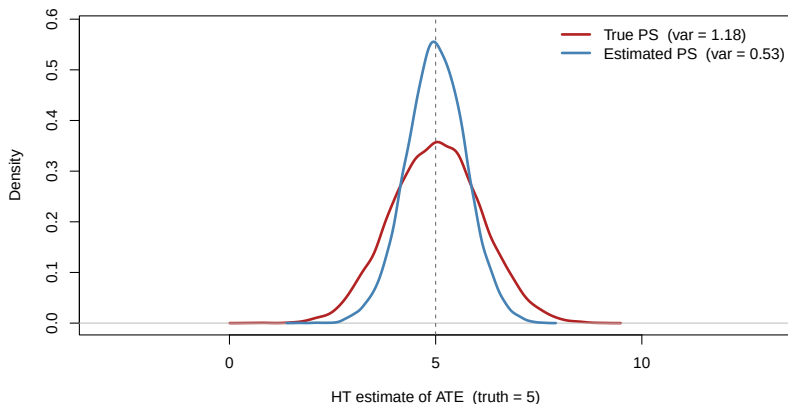
Monte Carlo: True vs Estimated Propensity Scores

- Toy DGP with one binary covariate X , true $\pi(X) = 0.5X + 0.4(1 - X)$, $Y = 5D - 10X + \mathcal{N}(0, 25)$. True ATE = 5.

```
1 > ht.est <- function(y, d, w) (1/length(y)) * sum(y*d*w - y*(1-d)*w)
2 > sims <- 10000; n <- 200
3 > true.holder <- est.holder <- rep(NA, sims)
4 > for (i in 1:sims) {
5 +   x <- rbinom(n, 1, 0.5)
6 +   dprobs <- 0.5*x + 0.4*(1-x)
7 +   d <- rbinom(n, 1, dprobs)
8 +   y <- 5*d - 10*x + rnorm(n, sd = 5)
9 +
10 +   true.w <- ifelse(d == 1, 1/dprobs, 1/(1-dprobs)) # uses TRUE pi
11 +   pprobs <- predict(glm(d ~ x)) # estimate pi from data
12 +   est.w <- ifelse(d == 1, 1/pprobs, 1/(1-pprobs))
13 +
14 +   true.holder[i] <- ht.est(y, d, true.w)
15 +   est.holder[i] <- ht.est(y, d, est.w)
16 + }
```

- Same HT estimator, same data — the only difference is whether weights use the *true* or the *estimated* propensity score.

Sampling Distribution of $\hat{\tau}_{HT}$



- Both estimators are centered at the truth (dashed line at $\tau = 5$): both are consistent.
- The *estimated- $\hat{\pi}$* sampling distribution is visibly tighter, with about half the variance of the true- π version.

Why Use *Estimated* Propensity Scores?

- A counterintuitive result: weighting by the estimated $\hat{\pi}(\mathbf{X})$ has *smaller* asymptotic variance than weighting by the true $\pi(\mathbf{X})$ (Hirano, Imbens, and Ridder, 2003).
- Intuition. The true π corrects only the *expected* imbalance between treated and control groups. The estimated $\hat{\pi}$, fit on the actual sample, also absorbs the *chance* imbalance that arose in this realization.
- Familiar parallel: in an RCT, OLS with covariate adjustment (Lin, 2013) is more efficient than the simple difference-in-means, even though randomization makes the difference unbiased.
- Caveat: the gain depends on the PS model being correctly specified, or estimated flexibly enough (HIR use sieve estimation).

Augmented IPW: Doubly Robust

- **Augmented** IPW (AIPW) combines regression and weighting:

$$\hat{\tau}_{\text{aipw}} = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i) + \left(\frac{D_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{\pi}(\mathbf{X}_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1 - \hat{\pi}(\mathbf{X}_i)} \right) \right\}$$

- **Double robustness:** consistent if either $\hat{\pi}(\mathbf{X}_i)$ or $\hat{\mu}_d(\mathbf{X}_i)$ is consistent.
 - Can allow each model to be more flexible without hurting asymptotics.
- **Efficient:** lowest asymptotic variance among consistent estimators when the PS model is correct.
- Bottom line: AIPW already jointly handles both the $\mathbf{X} \rightarrow D$ nuisance and the $\mathbf{X} \rightarrow Y$ nuisance, a 1994 idea (Robins, Rotnitzky and colleagues). Remember this for the DML wrap-up.

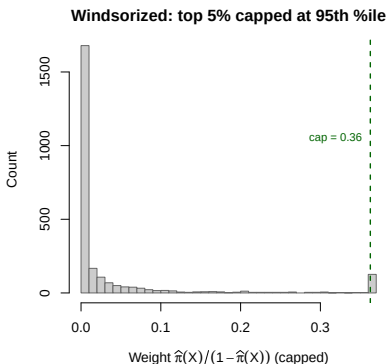
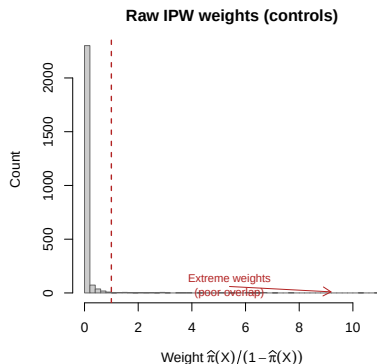
Generating IPW Weights for the ATT (LaLonde)

- Same lalonde_ECI data as the matching demo; now re-weight controls instead of pruning them.

```
1 # Generating propensity score weights for the ATT
2 > library(WeightIt)
3 > W.out <- WeightIt::weightit(
4   treat ~ age + I(age^2) + black + married + hispanic + re75,
5   data = lalonde_ECI,
6   method = "ps",          # logistic regression for pscore
7   estimand = "ATT"        # weights controls by pi(X)/(1 - pi(X))
8 )
9 > summary(W.out)
```

- Same covariates as the matching PS specification, so matching ATT and weighting ATT are directly comparable.
- With estimand = "ATT": treated units keep weight 1; controls are weighted by $\hat{\pi}(\mathbf{X})/(1 - \hat{\pi}(\mathbf{X}))$.

Inspecting the Estimated Weights



- Left: most controls have $w \approx 0$, but a long right tail beyond $w = 1$ (red dashed) reaches ≈ 10 . These come from controls with $\hat{\pi}(X) \approx 1$: regions of poor overlap.
- Right: Windsorize the top 5% (cap w at the 95th percentile). The bump at the right edge is the capped mass.
- Note: Hajek normalization (*previous slide*) only *rescales* weights; it does not change the shape. The variance benefit shows up in the *estimator*, not in the weight distribution.

Computing the ATT: Horvitz-Thompson and Hajek

- Both estimators in a single function, ready for the bootstrap below:

```
1  ## append estimated weights
2  > lalonde_ECI <- lalonde_ECI %>% mutate(weights = W.out$weights)
3
4  ## ATT (returns both HT and Hajek estimates)
5  > att_ipw <- function(dat, indices = NULL) {
6    if (is.null(indices)) indices <- 1:nrow(dat)
7    dat <- dat[indices, ]
8
9    w <- dat %>% filter(treat == 0) %>% pull(weights)
10   rew <- w / sum(w) # normalized weights
11   Y1 <- dat %>% filter(treat == 1) %>% pull(re78)
12   Y0 <- dat %>% filter(treat == 0) %>% pull(re78)
13   n1 <- length(Y1)
14
15   att_ht <- (sum(Y1) - sum(Y0 * w)) / n1 # Horvitz-Thompson ATT
16   att_hjk <- mean(Y1) - sum(rew * Y0) # Hajek ATT (normalized)
17   c(HT = att_ht, Hajek = att_hjk)
18 }
19 > att_ipw(lalonde_ECI)
```

- HT divides by n_1 and uses raw weights; Hajek normalizes weights to sum to 1, which makes it less sensitive to extreme draws.

Adding the Outcome Model: AIPW

- Augment IPW with an outcome model $\hat{\mu}_0(\mathbf{X})$ for the treated counterfactual. **Doubly robust**: consistent if *either* $\hat{\pi}$ or $\hat{\mu}_0$ is correct (Robins, Rotnitzky, and Zhao, 1994).

```
1 > att_aipw <- function(dat, indices = NULL) {
2   if (is.null(indices)) indices <- 1:nrow(dat)
3   d <- dat[indices, ]
4
5   pi_hat <- predict(
6     glm(treat ~ age + I(age^2) + black + married + hispanic + re75,
7       family = binomial, data = d), type = "response")
8   mu_0 <- predict(
9     lm(re78 ~ age + I(age^2) + black + married + hispanic + re75,
10      data = filter(d, treat == 0)), newdata = d)
11
12   resid <- d$re78 - mu_0
13   n1 <- sum(d$treat == 1)
14   (sum(d$treat * resid) -
15    sum((1 - d$treat) * pi_hat / (1 - pi_hat) * resid)) / n1
16 }
17 > att_aipw(lalonde_ECI)
```

- ATT only needs $\hat{\mu}_0$ (impute $Y(0)$ for the treated); ATE form would need $\hat{\mu}_1$ too.

Standard Errors via the Bootstrap

- Easiest route to SE with estimated $\hat{\pi}$ (and $\hat{\mu}_0$); avoids hand-derived delta method.

```
1 > library(boot)
2 > set.seed(42)
3 > b_ipw <- boot(data = lalonde_ECI, statistic = att_ipw, R = 1000)
4 > b_aipw <- boot(data = lalonde_ECI, statistic = att_aipw, R = 1000)
5
6 > data.frame(
7   estimator = c("HT", "Hajek", "AIPW"),
8   point      = c(b_ipw$t0,      b_aipw$t0),
9   boot_SE    = c(apply(b_ipw$t, 2, sd), sd(b_aipw$t))
10 )
11 #>   estimator   point   boot_SE
12 #> 1      HT <pt_HT> <SE_HT>    # raw weights -> large SE
13 #> 2   Hajek <pt_Hjk> <SE_Hjk>    # normalized -> smaller SE
14 #> 3    AIPW <pt_AIPW> <SE_AIPW>    # outcome model absorbs noise
```

- Expect: $SE_HT \gg SE_Hjk > SE_AIPW$. The outcome model in AIPW soaks up additional variance.
- Compare with the matching estimate (Abadie-Imbens SE, $\hat{\tau} \approx -941$). Different point estimates plus much wider CIs in IPW are the data telling us about poor overlap.

Problems with Weighting

- Propensity score balances covariates across **repeated samples**
 - Relies on the law of large numbers (not optimized for small samples).
- Difficult to model the propensity score for covariate balance.
- Highly variable/unstable weights $\rightsquigarrow$ high variance estimators.
 - When there is a lack of overlap so $\pi(\mathbf{X}_i)$ is close to 0 or 1.
 - **Windsorizing**: trim weights beyond e.g. the 5th and 95th percentile.

Covariate Balancing Propensity Scores (CBPS)

- How to help with these issues: prioritize balance in the estimation of $\pi(\mathbf{X}_i)$.
 - One approach: **covariate balancing propensity scores** (Imai and Ratkovic).
- Usual MLE solves a score moment condition:

$$\mathbb{E} \left[\underbrace{\left(\frac{D_i}{\pi_{\theta}(\mathbf{X}_i)} - \frac{1 - D_i}{1 - \pi_{\theta}(\mathbf{X}_i)} \right) \frac{\partial \pi_{\theta}(\mathbf{X}_i)}{\partial \theta}}_{\text{score}} \right] = 0$$

- CBPS *adds* balancing conditions:

$$\mathbb{E} \left[\left(\frac{D_i}{\pi_{\theta}(\mathbf{X}_i)} - \frac{1 - D_i}{1 - \pi_{\theta}(\mathbf{X}_i)} \right) f(\mathbf{X}_i) \right] = 0$$

- If $f(\mathbf{X}_i) = \mathbf{X}_i$: condition forces the means to balance.
 - If $f(\mathbf{X}_i) = \mathbf{X}_i^2$: balances the second moments.
- Estimation by GMM (more moments than parameters $\Rightarrow$ minimum-distance solution).

Balancing Weights (Zubizarreta, 2015)

- Alternative: find weights that *directly* balance covariates.
- **Stable balancing weights** as one recent example. Rearrange so first m units are controls.
- Solve the convex quadratic program:

$$\begin{aligned} & \min_{w_1, \dots, w_m} \sum_{i=1}^m (w_i - \bar{w})^2 \\ \text{s.t. } & \sum_{i=1}^m w_i = 1, \quad w_i \geq 0, \quad \left| \frac{1}{n_1} \sum_{i:D_i=1} X_{ik} - \sum_{i=1}^m w_i X_{ik} \right| \leq \delta_k \end{aligned}$$

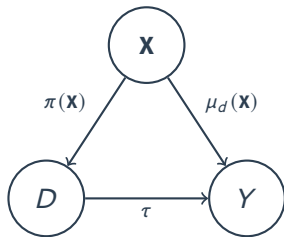
- Minimum-variance weights that *approximately* balance covariates.
 - δ_k : amount of allowed imbalance, chosen by the researcher.
 - Can include X_{ik}^2 etc. to balance other parts of the distribution.

2/ Looking Ahead: From AIPW to DML

The Whole Journey: Where Do Good Controls Come From?

- Through this course, every method has been one answer to the same question.
- **Good controls come from:**
 - **RCT:** units *randomized* into control.
 - **SOO:** units with *similar X* (if we accept conditional unconfoundedness).
 - **IV:** units with *opposite Z*.
 - **DiD / Panel:** a unit's own past, i.e., variation *within* a unit over time.
 - **RDD** (next week): units *near a cutoff*, in the limit.
- Today, within the SOO branch, we tightened the comparison two ways:
 - **Matching:** prune controls down to comparable ones (Gary King's "pruning").
 - **Weighting:** keep all controls, re-weight them to emulate the missing population (HT logic, applied to treatment).

Three Views of the Same DAG: Matching, RA, AIPW



Two nuisance edges to deal with:

- $\mathbf{X} \rightarrow D$: *treatment model* $\pi(\mathbf{X})$
- $\mathbf{X} \rightarrow Y$: *outcome model* $\mu_d(\mathbf{X})$

Three lenses on the same problem:

- **Matching / IPW**: tackles only $\mathbf{X} \rightarrow D$.
- **Reg. adjustment**: tackles only $\mathbf{X} \rightarrow Y$.
The Freedman/Lin debate is about *how* (uninteracted vs. fully interacted); the target is the same edge.
- **AIPW** (Robins et al., 1994): handles *both*, i.e., doubly robust.

↪ AIPW already gives us both edges. So what is new in 2018?

So What Is New in 2018?

- AIPW = doubly robust treatment of both nuisances. A **1990s** idea (Robins, Rotnitzky, Zhao).
- **Chernozhukov et al. (2018)**: “Double/Debiased Machine Learning”
 - The new ingredient: use **machine learning** to estimate the nuisances $\widehat{\pi}(\mathbf{X}), \widehat{\mu}_d(\mathbf{X})$.
 - Why bother? high-dimensional $\mathbf{X}$, unknown functional form, less reliance on parametric guesses.
- But: naïvely plugging ML into AIPW introduces *two new biases* that didn't exist with logistic regression:
 - **Regularization bias**: LASSO/RF penalties shrink $\widehat{\pi}, \widehat{\mu}$, and that bias leaks into $\widehat{\tau}$.
 - **Overfitting bias**: the same data are used for both nuisance estimation and $\widehat{\tau}$, producing spurious correlations.

The Two Pillars of DML

- **Pillar 1: Neyman-orthogonal score.**
 - Construct a moment condition whose derivative with respect to the nuisance parameters vanishes at the truth.
 - Consequence: $\hat{\tau}$ is first-order **insensitive** to plug-in error from regularization.
 - The AIPW score from the previous frame is already Neyman-orthogonal.
- **Pillar 2: Cross-fitting.**
 - Train $\hat{\pi}, \hat{\mu}$ on one fold; evaluate $\hat{\tau}$ on the held-out fold; rotate and average.
 - Kills the overfitting bias.
- Together: $\sqrt{n}$ -consistent, asymptotically normal $\hat{\tau}$ even with off-the-shelf ML for $\hat{\pi}, \hat{\mu}$.
- Today's matching and weighting machinery is exactly where that line of work picks up; a natural direction for further reading once you are comfortable with the AIPW score.



On to the Presentations & Discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>

